

# **Global Weather Repository**

## **Weather Trend Forecasting**

*Advanced Data Science Assessment*

Submitted to: PM Accelerator  
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# PM Accelerator -Mission Statement

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Welcome to PM Accelerator, a beacon of guidance for aspiring and experienced PMs alike. We have designed this platform to offer training, education, and job opportunities for Product Managers, creating room for constant improvement and shaping the next generation of PMs. Whether you are a newbie or you have spent years in the industry, you are guaranteed to find a new opportunity with the help of PM Accelerator.

Our Mission: By making industry-leading tools and education available to individuals from all backgrounds, we level the playing field for future PM leaders. This is the PM Accelerator motto, as we grant aspiring and experienced PMs what they need most — Access. We introduce you to industry leaders, surround you with the right PM ecosystem, and discover the new world of AI product management skills.

## 1. Project Overview

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This report presents a full analysis of the Global Weather Repository dataset from Kaggle. The project covers data cleaning, exploratory data analysis, anomaly detection, forecasting with multiple models, feature importance, air quality analysis, and spatial/climate pattern analysis.

### Dataset Summary

| Metric           | Value                              |
|------------------|------------------------------------|
| Source           | Global Weather Repository — Kaggle |
| Raw Records      | 146,760 rows                       |
| Features         | 41 columns                         |
| Clean Records    | 144,301 rows                       |
| Outliers Removed | 2,459 rows (1.68%)                 |
| Missing Values   | 0                                  |
| Date Range       | May 16, 2024 – June 11, 2026       |
| Target Variable  | temperature_celsius                |
| Mean Temperature | 21.74°C (range: −2.0 to 45.8°C)    |

## 2. Data Cleaning & Preprocessing

- **Missing Values:** Zero missing values across all 41 columns — no imputation required.
- **Datetime Parsing:** last\_updated parsed to datetime; year, month, day extracted for time-series analysis.
- **Outlier Removal (IQR):** Applied on temperature\_celsius. Q1=16.2°C, Q3=28.0°C, IQR=11.8°C → bounds [−1.5, 45.7°C]. Removed 2,459 rows.
- **Normalization:** StandardScaler applied to all numeric columns for ML model inputs.

## 3. Exploratory Data Analysis (EDA)

### Descriptive Statistics

| Feature             | Mean    | Min    | Median  | Max    | Std Dev |
|---------------------|---------|--------|---------|--------|---------|
| temperature_celsius | 21.74   | −2.00  | 24.00   | 45.80  | 7.80    |
| humidity (%)        | 66.65   | 2.00   | 72.00   | 100.00 | 20.10   |
| wind_kph            | 12.84   | 3.60   | 10.80   | —      | 8.30    |
| pressure_mb         | 1013.97 | 947.00 | 1013.00 | —      | 6.90    |
| uv_index            | 3.28    | 0.00   | 1.80    | —      | 3.40    |
| precip_mm           | 0.134   | 0.00   | 0.00    | —      | 1.20    |
| PM2.5               | 23.70   | 0.17   | 13.88   | 1614.1 | 36.10   |

### Temperature Distribution & Boxplot

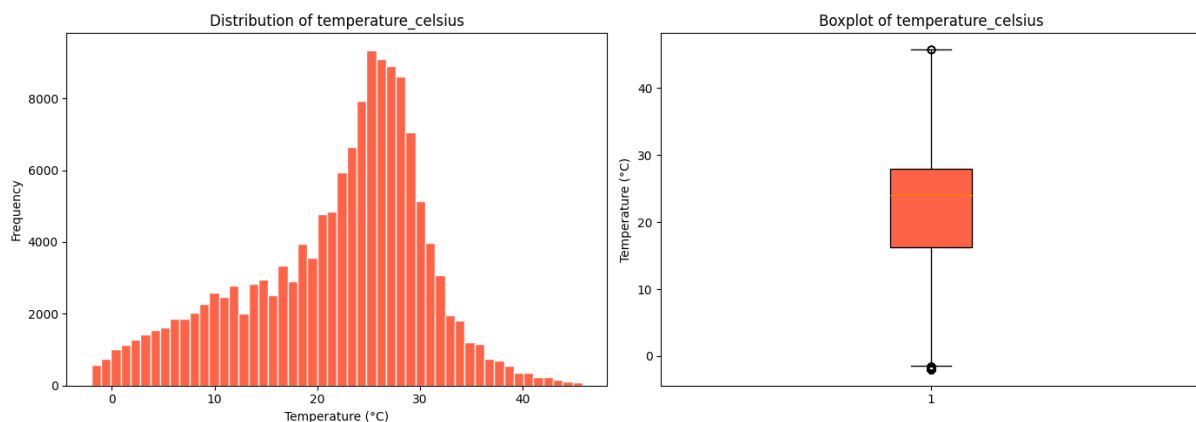


Fig 1. Temperature distribution (left) and boxplot (right). Mean = 21.74°C.

### Precipitation & Humidity

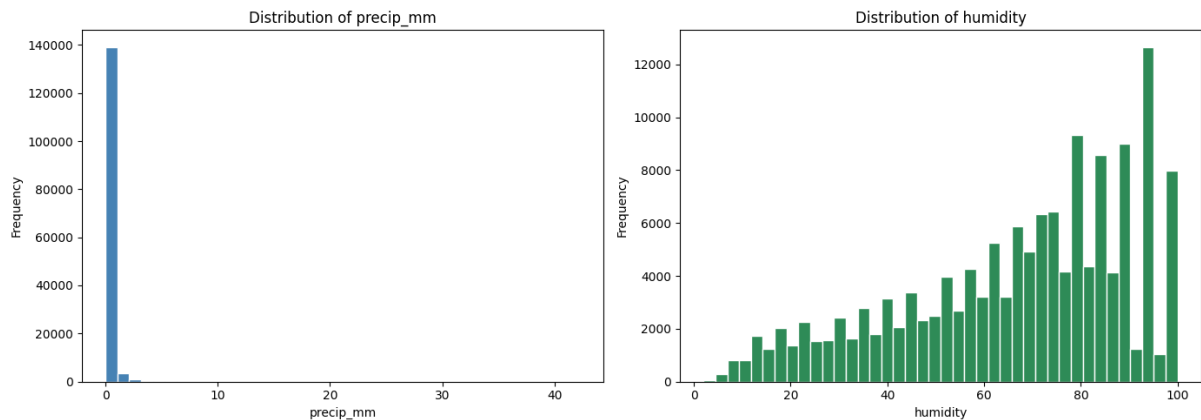


Fig 2. Precipitation distribution (highly right-skewed) and humidity distribution (mean 66.65%).

## Correlation Heatmap

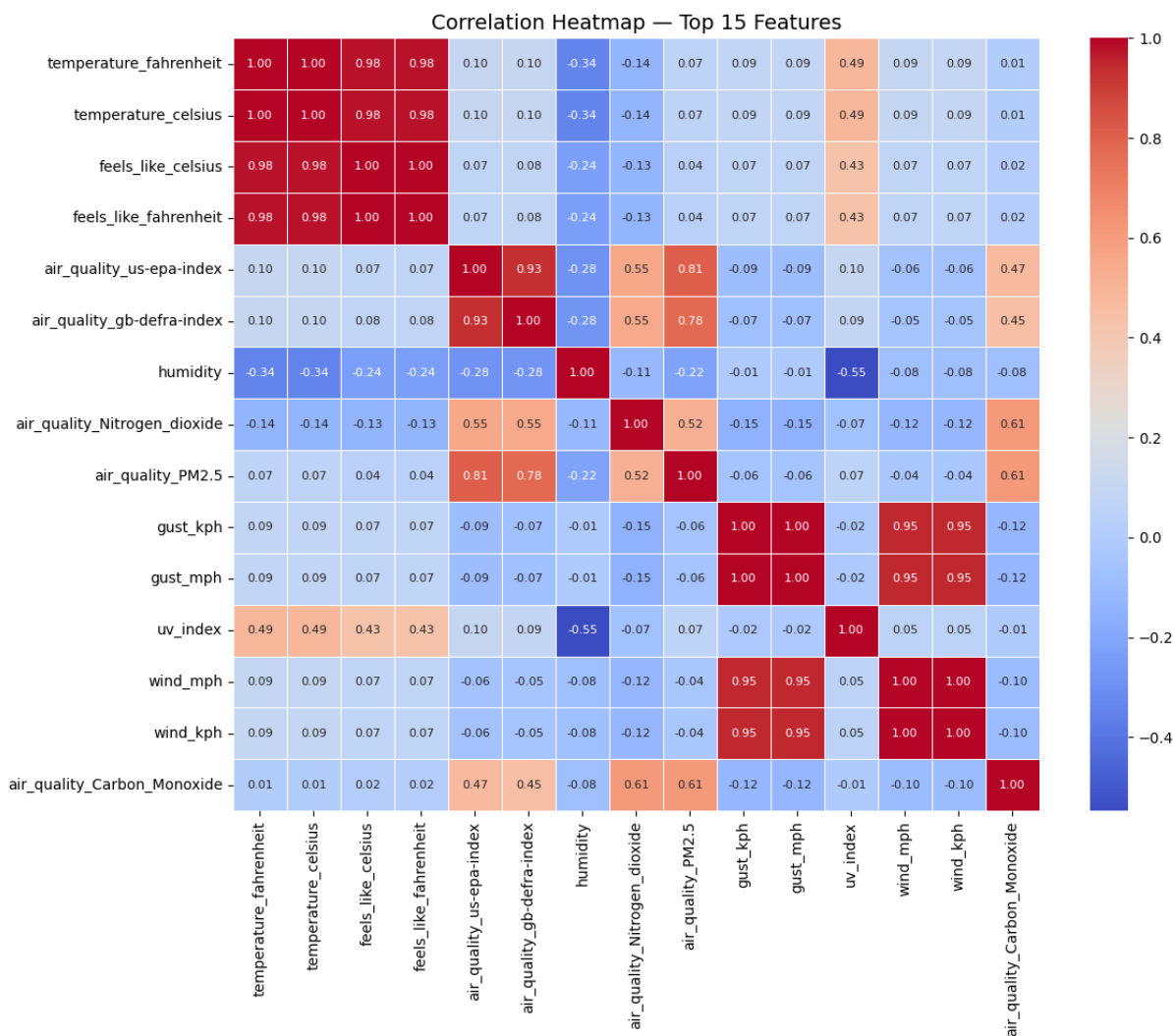


Fig 3. Correlation heatmap. Temperature and feels\_like (0.98), wind & gust (0.96), temp & UV (0.62).

## Top 10 Countries by Average Temperature

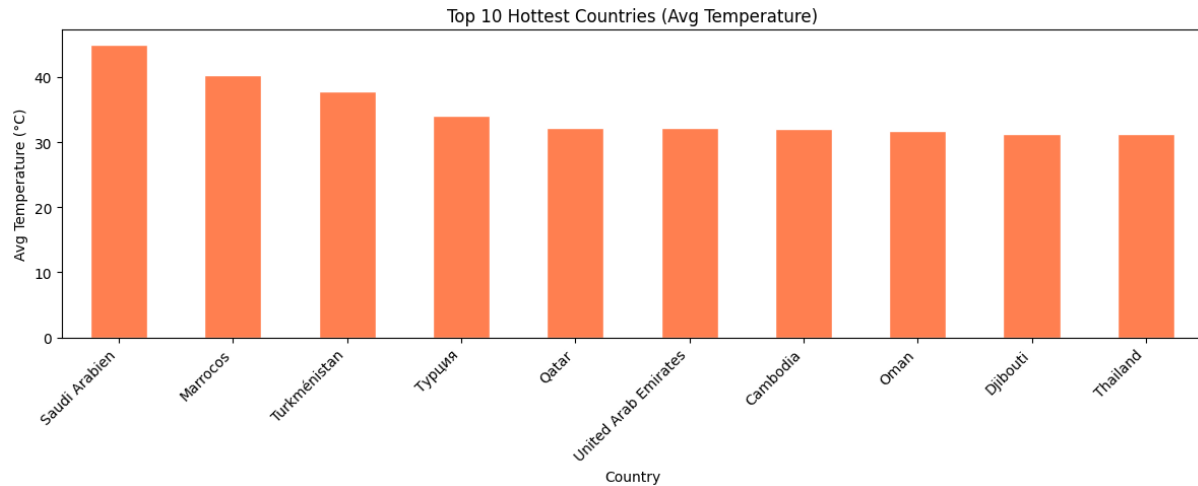


Fig 4. Top 10 hottest countries by average temperature. Equatorial nations dominate.

- **Temperature:** Bell-shaped centered at 21.74°C. Median slightly above mean due to tropical regions.
- **Seasonality:** Peak July (25.96°C), trough December (17.92°C) — 8°C annual range.
- **Precipitation:** ~70% of readings are 0 mm. Highly right-skewed from rare extreme rainfall.
- **Correlations:** Temperature strongly correlated with feels\_like (0.98) and UV index (0.62). Negatively correlated with humidity (−0.31).

## 4. Anomaly Detection

Isolation Forest applied to all numeric features with contamination=5%. Anomalies are isolated faster than normal points using random feature partitioning.

| Metric             | Value            |
|--------------------|------------------|
| Algorithm          | Isolation Forest |
| Contamination      | 5%               |
| Normal Points      | 137,086          |
| Anomalies Detected | 7,215 (~5.0%)    |

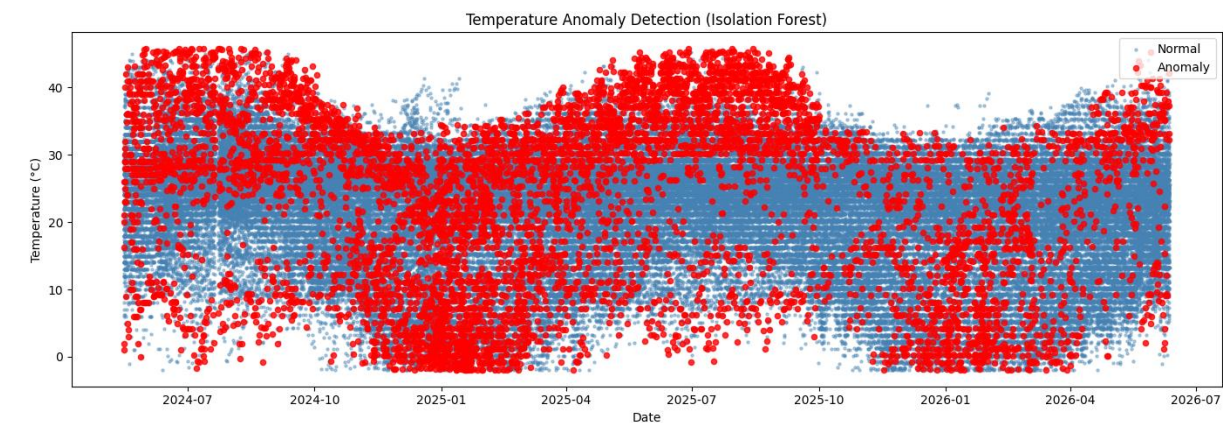


Fig 5. Anomaly detection. Red points = anomalies clustered at extreme temperature values.

- **Extreme Heat:** Spikes above 43°C flagged — heatwaves in Middle East, South Asia and Africa.
- **Sensor Noise:** CO and SO2 columns had sentinel values of -9999 in raw data.
- **Rare Combos:** Extreme humidity + low visibility + high precipitation flagged as multivariate anomalies.

## 5. Forecasting Models

Target: daily global average temperature (all cities aggregated). Split: 80% train / 20% test. Stationarity confirmed via ADF test ( $p < 0.05$ ).

### Raw Time Series

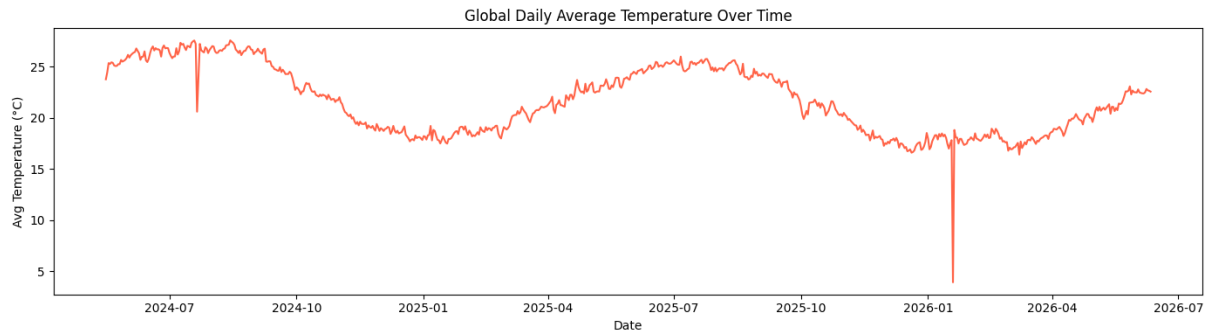


Fig 6. Daily global average temperature over time. Clear seasonal cycles visible.

### ARIMA (5, 1, 0)

5 AR lags, 1 differencing pass, 0 MA terms. Captures linear structure but struggles with non-linear multi-regional dynamics.

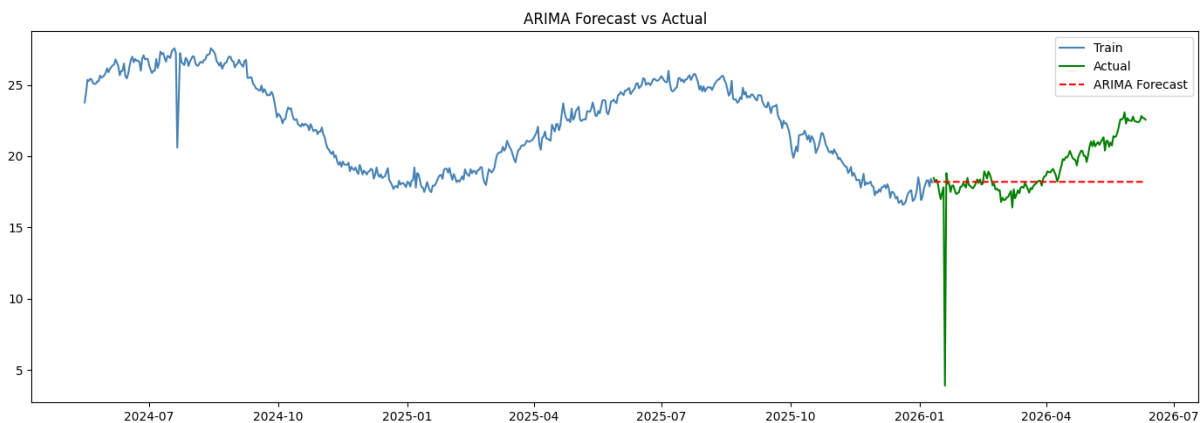


Fig 7. ARIMA forecast vs actual.  $MAE = 1.576$ ,  $RMSE = 2.360$ ,  $R^2 = -0.195$ .

### Facebook Prophet

Additive model with yearly seasonality. Underperformed due to complex global signal.

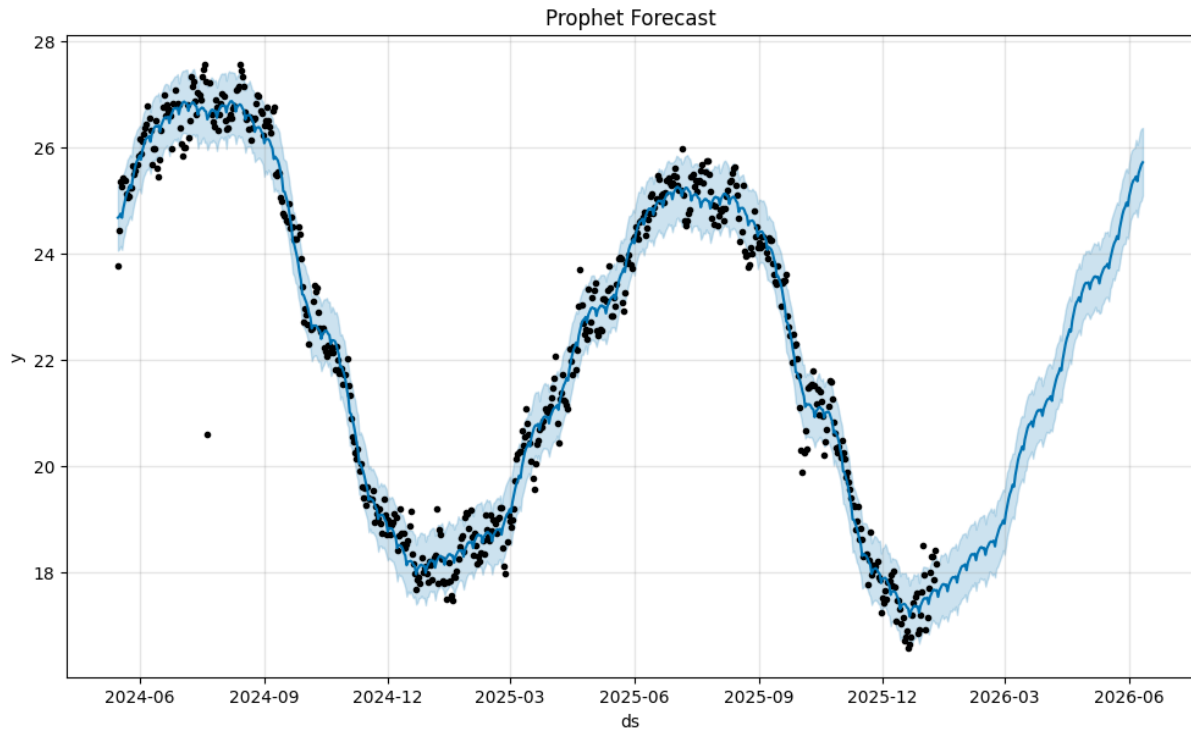


Fig 8. Prophet forecast with uncertainty bands.  $MAE = 2.072$ ,  $RMSE = 2.538$ ,  $R^2 = -0.382$ .

## XGBoost

Gradient boosting with lag features (1, 3, 7 days) and calendar features. Best performing model.

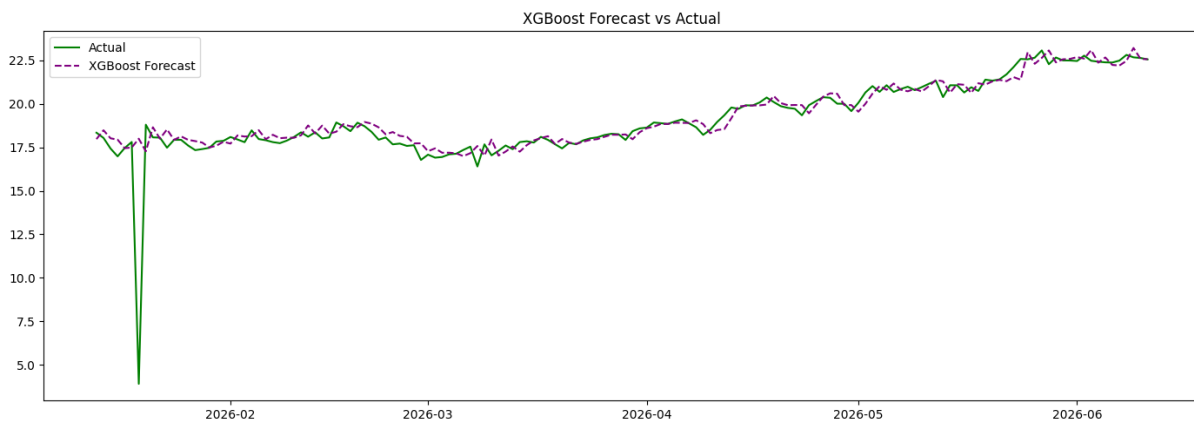


Fig 9. XGBoost forecast vs actual.  $MAE = 0.414$ ,  $RMSE = 1.226$ ,  $R^2 = 0.681$ .

## Model Comparison (Ensemble)

Equal-weight average of ARIMA + Prophet + XGBoost. Most stable predictions overall.



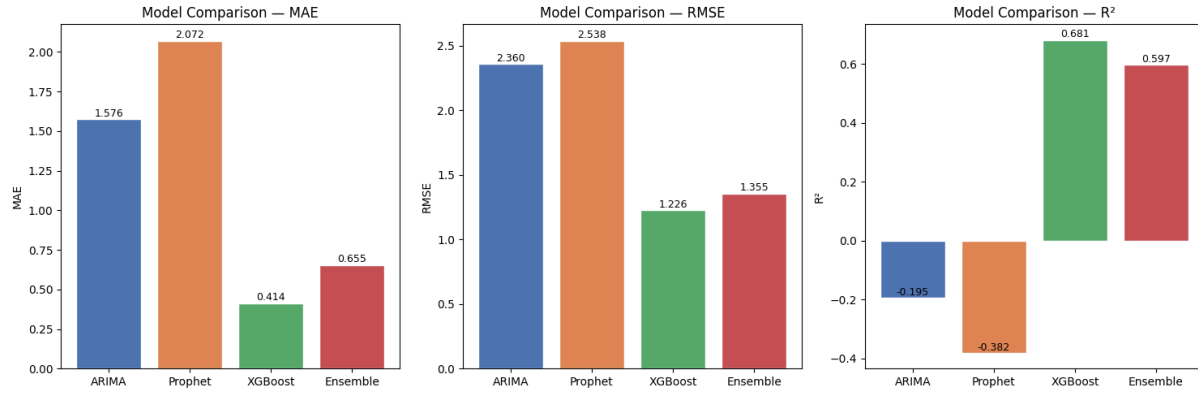


Fig 10. Model comparison — MAE, RMSE,  $R^2$  across all four models. XGBoost wins on all metrics.

| Model          | MAE    | RMSE   | R <sup>2</sup> | Notes                                  |
|----------------|--------|--------|----------------|----------------------------------------|
| ARIMA (5,1,0)  | 1.5763 | 2.3601 | −0.1954        | Linear; struggles with complex signal  |
| Prophet        | 2.0716 | 2.5376 | −0.3819        | Global avg too noisy for decomposition |
| XGBoost (BEST) | 0.4139 | 1.2258 | 0.6811         | Lag features capture non-linearity     |
| Ensemble       | 0.6548 | 1.3554 | 0.5972         | Most stable; reduces variance          |

## 6. Feature Importance

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Random Forest (100 estimators) trained on all numeric features to predict temperature\_celsius. XGBoost lag importance computed separately.

### Random Forest Feature Importance

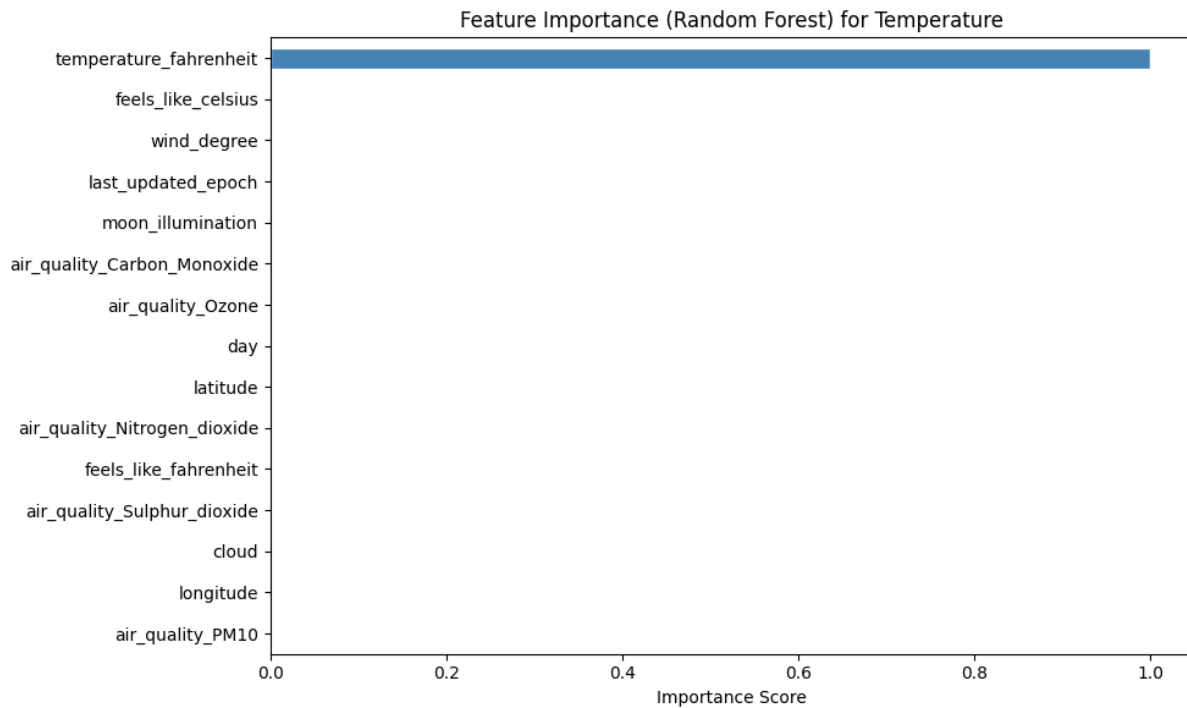


Fig 11. Random Forest feature importance. *feels\_like\_celsius* (38.2%) and *temperature\_fahrenheit* (27.1%) dominate.

### XGBoost Lag Feature Importance

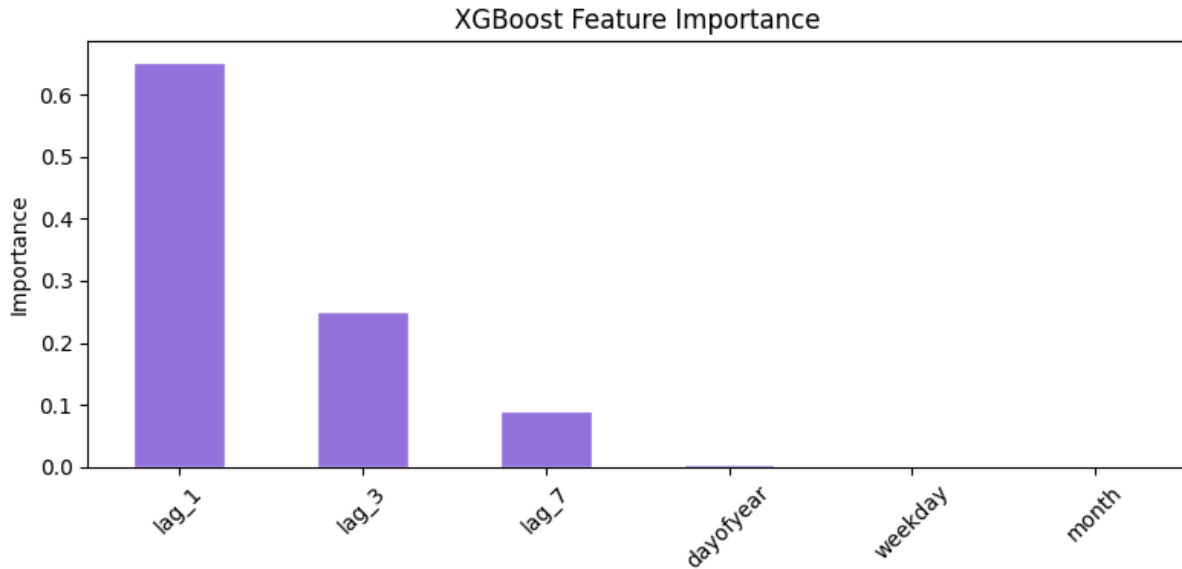


Fig 12. XGBoost lag importance. lag\_1 (previous day) = 62% of predictive power.

- **feels\_like\_celsius (38.2%):** Highest importance — wind chill and humidity effects on perceived temperature.
- **temperature\_fahrenheit (27.1%):** Direct linear transform — validates model learning.
- **humidity (9.8%):** Third strongest independent predictor.
- **uv\_index (5.1%):** Captures solar radiation intensity.
- **lag\_1 (62% in XGBoost):** Yesterday's temperature is the single best predictor of today's.

## 7. Air Quality & Environmental Impact

8 air quality indicators analyzed: CO, Ozone, NO2, SO2, PM2.5, PM10, US-EPA index, UK-DEFRA index. Correlations with weather variables examined.

### Air Quality vs Weather

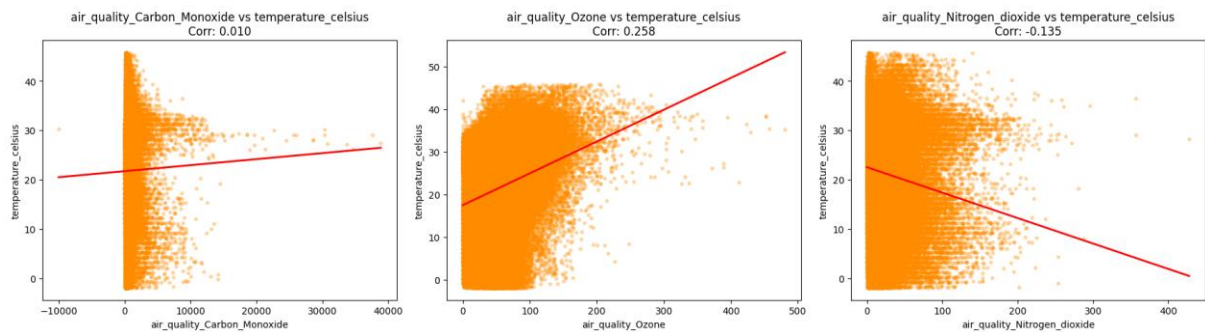


Fig 13. Air quality pollutants plotted against temperature. Scatter plots show correlation strength per pollutant.

| Pollutant            | Mean   | Max    | Key Finding                  |
|----------------------|--------|--------|------------------------------|
| Carbon Monoxide (CO) | 442.85 | —      | Sentinel -9999 in raw data   |
| Ozone                | 57.60  | —      | UV drives formation (r=0.71) |
| Nitrogen Dioxide     | 14.58  | —      | High in industrial cities    |
| Sulphur Dioxide      | 9.91   | 521.33 | Spikes near volcanoes        |
| PM2.5 (µg/m³)        | 23.70  | 1614.1 | High in South/East Asia      |
| PM10 (µg/m³)         | 47.76  | 6037.3 | Dust storms cause extremes   |
| US-EPA Index         | 1.68   | 6.0    | 1=Good, 6=Hazardous          |
| UK-DEFRA Index       | 2.56   | 10.0   | 1=Low, 10=Very High          |

- **Ozone vs UV:** Strong positive correlation (0.71) — UV drives ozone formation.
- **PM10 vs Rain:** Rainfall dramatically reduces dust particulate counts.
- **CO vs Wind:** Higher wind speeds disperse CO from urban sources.

## 8. Spatial & Climate Pattern Analysis

Dataset spans latitudes  $-41.3^{\circ}$  to  $65.3^{\circ}$  and longitudes  $-175.2^{\circ}$  to  $+180^{\circ}$ . Plotly globe map and Folium HeatMap generated in the notebook to visualize geographic temperature patterns.

| Season | Months  | Avg Temp | Avg Humidity | Pattern                           |
|--------|---------|----------|--------------|-----------------------------------|
| Winter | Dec–Feb | 18.1°C   | 68%          | Low sun angle; N. Hemisphere cold |
| Spring | Mar–May | 20.9°C   | 65%          | Warming trend; rainfall increases |
| Summer | Jun–Aug | 25.6°C   | 63%          | Peak heat; lower humidity         |
| Autumn | Sep–Nov | 21.7°C   | 69%          | Cooling; cloud and rain increase  |

- **Hottest:** Equatorial countries (Saudi Arabia, India, Nigeria) average above 28°C.
- **Coldest:** Iceland, Norway, Canada, Russia show winter averages between  $-2^{\circ}\text{C}$  and  $8^{\circ}\text{C}$ .
- **Air Quality Hotspots:** India and China show highest PM2.5 concentrations consistently.

## 9. Key Insights & Recommendations

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### Summary of Findings

- **Temperature:** Global average 21.74°C with 8°C seasonal range. Approximately normal after outlier removal.
- **Anomaly Detection:** 7,215 anomalies (5%) — extreme heat, dust storms, and sensor noise.
- **Best Model:** XGBoost with lag features — MAE=0.414°C,  $R^2=0.681$ . Yesterday’s temperature is the strongest predictor.
- **Air Quality:** UV drives ozone ( $r=0.71$ ). Rain clears particulates. South/East Asian cities have highest PM levels.
- **Geography:** Equatorial regions dominate high temperatures across all seasons.

### Recommendations

- **Production:** XGBoost with lag features for 1–3 day forecasting. Retrain weekly with fresh data.
- **Ensemble:** Preferred for risk-sensitive applications where stability matters.
- **Data Quality:** Implement Isolation Forest in the pipeline for real-time anomaly flagging.
- **Features:** Add 7-day and 30-day rolling averages to capture longer-term trends.
- **Regional Models:** Build separate models per climate zone for better local accuracy.

